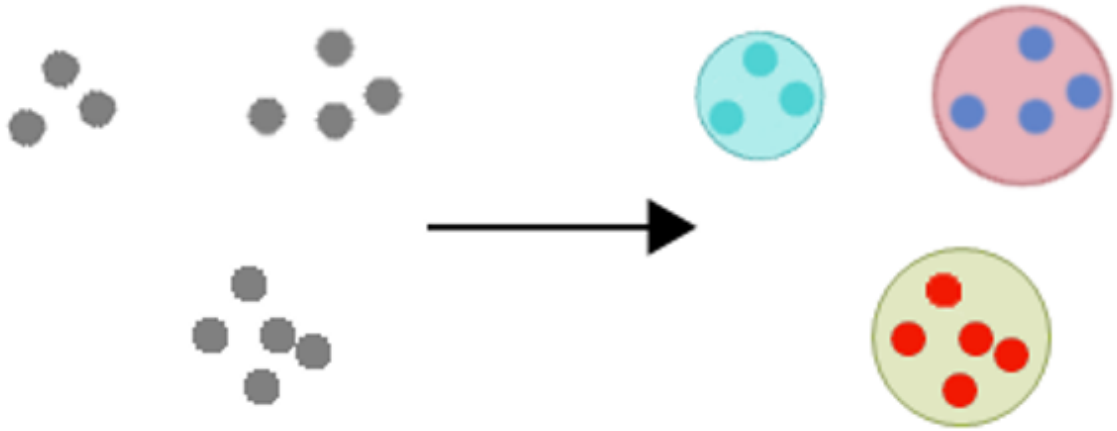


Figure 5.13 An example of clustering. On the left, we have a raw dataset of 2D points. We have no extra information about the points or the relationships between them. After clustering the dataset, on the right we can see that we have *inferred* some relationships between the points, and in particular we grouped them in 3 subsets whose points seems to show higher correlation.

Agglomerative hierarchical clustering starts with each point in its own *cluster* (partition) and continuously merges two points (and their clusters) till all clusters are merged into one. The algorithm keeps a history of this process, and it is possible to get a snapshot of the clusters created at any of the steps: the exact point where this snapshot is taken is controlled by a few parameters, and determines the result of the algorithm.